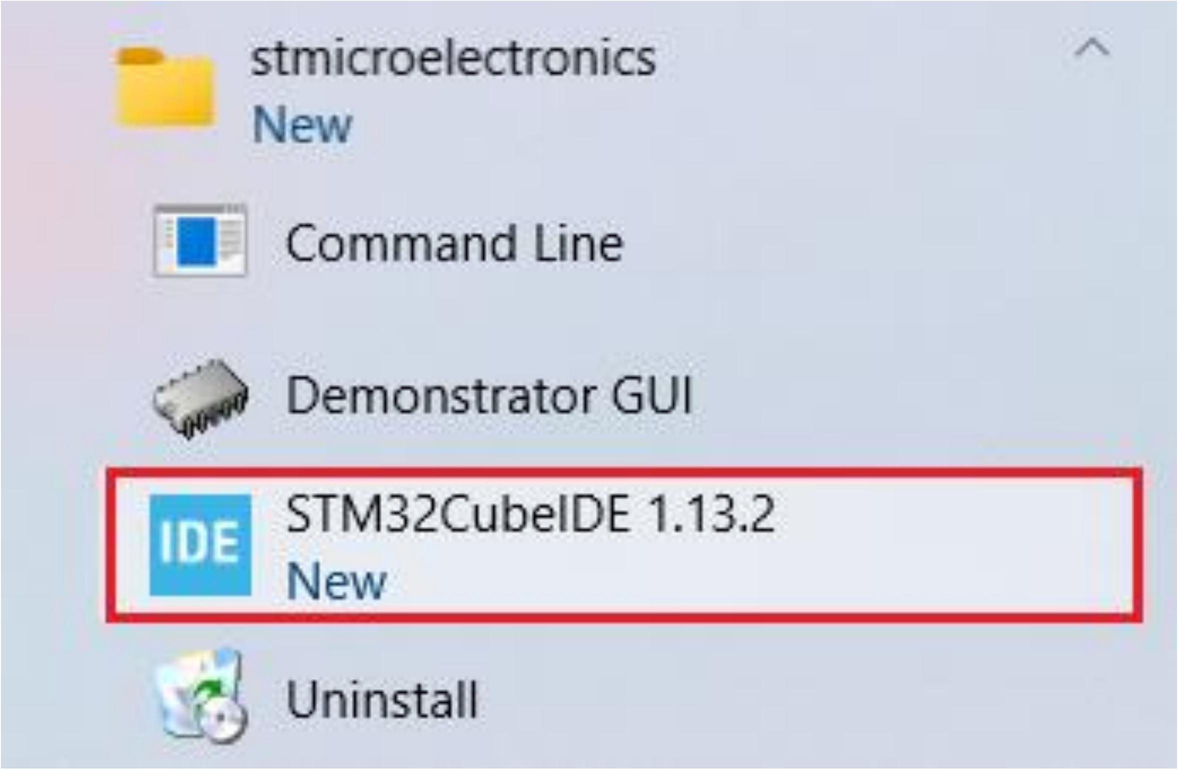
UART1 테스트

CRZ Technology

Mango-M32F2 UART1 테스트

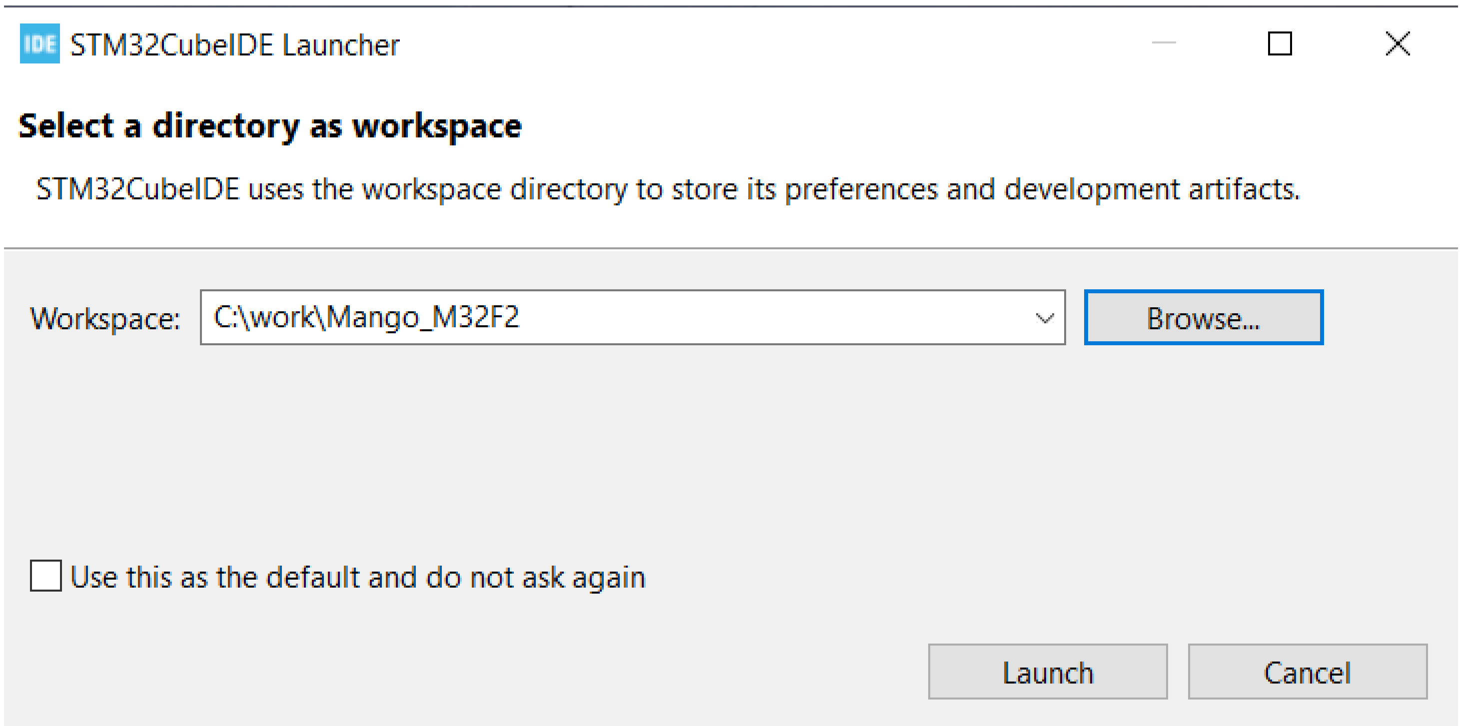
STM32CubeIDE 프로젝트 생성

■ **STM32CubeIDE 실행.**

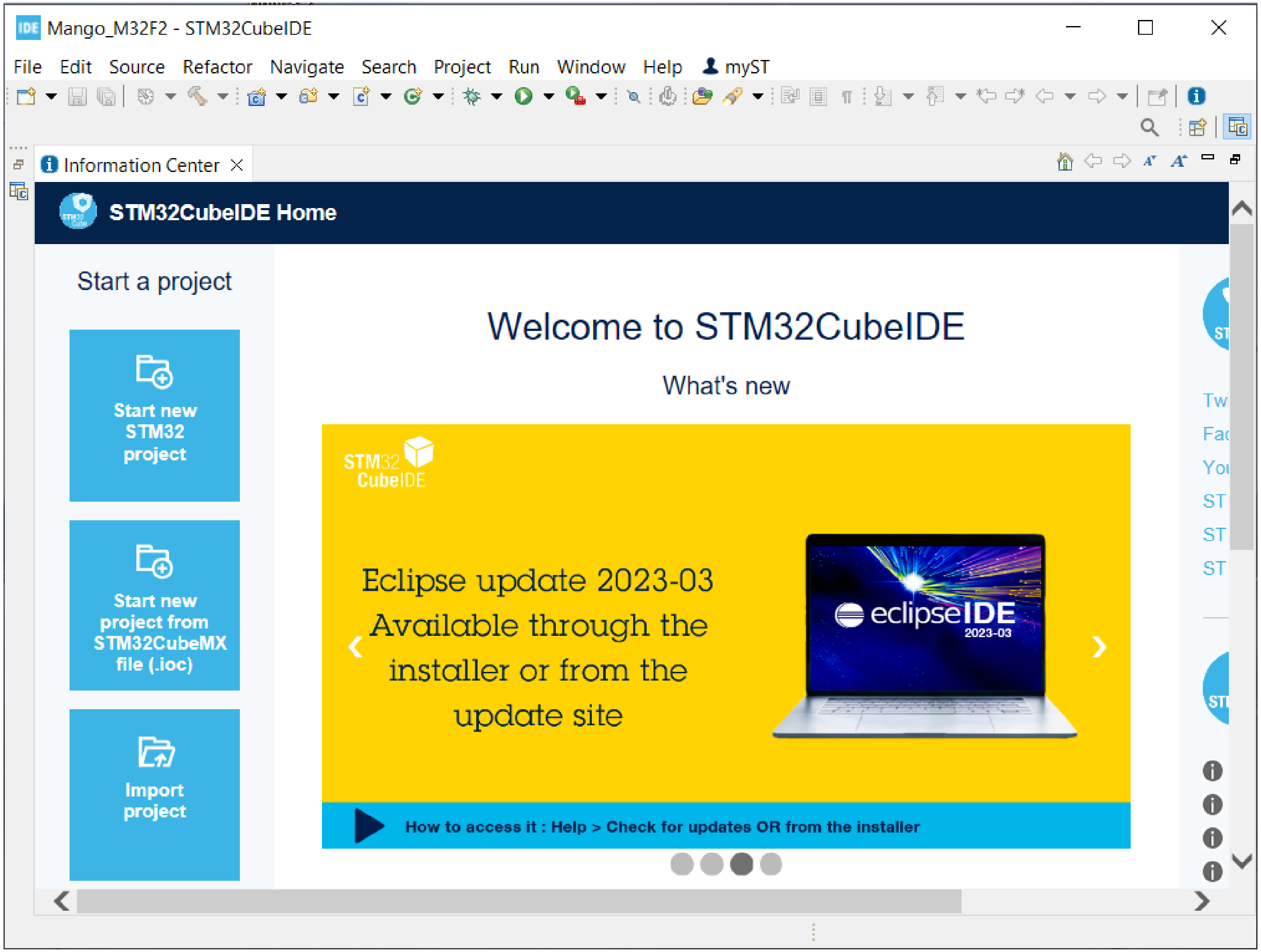


STM32CubeIDE 프로젝트 생성

workspace 설정

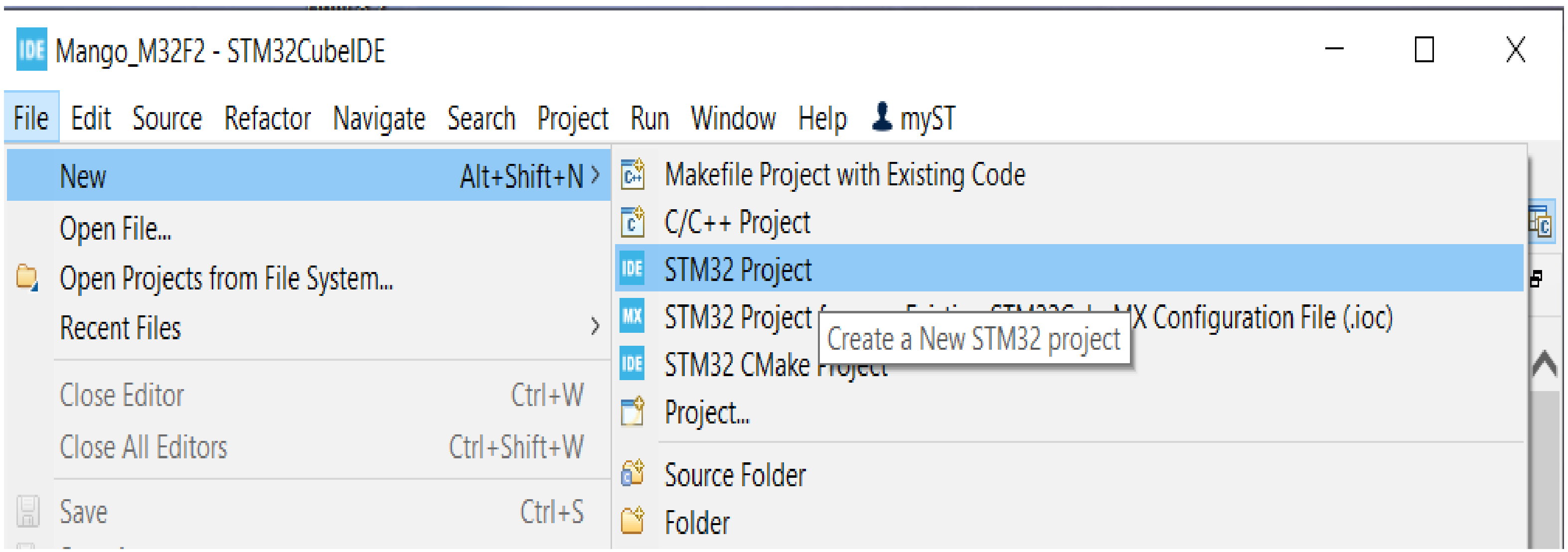


STM32CubeIDE 프로젝트 생성



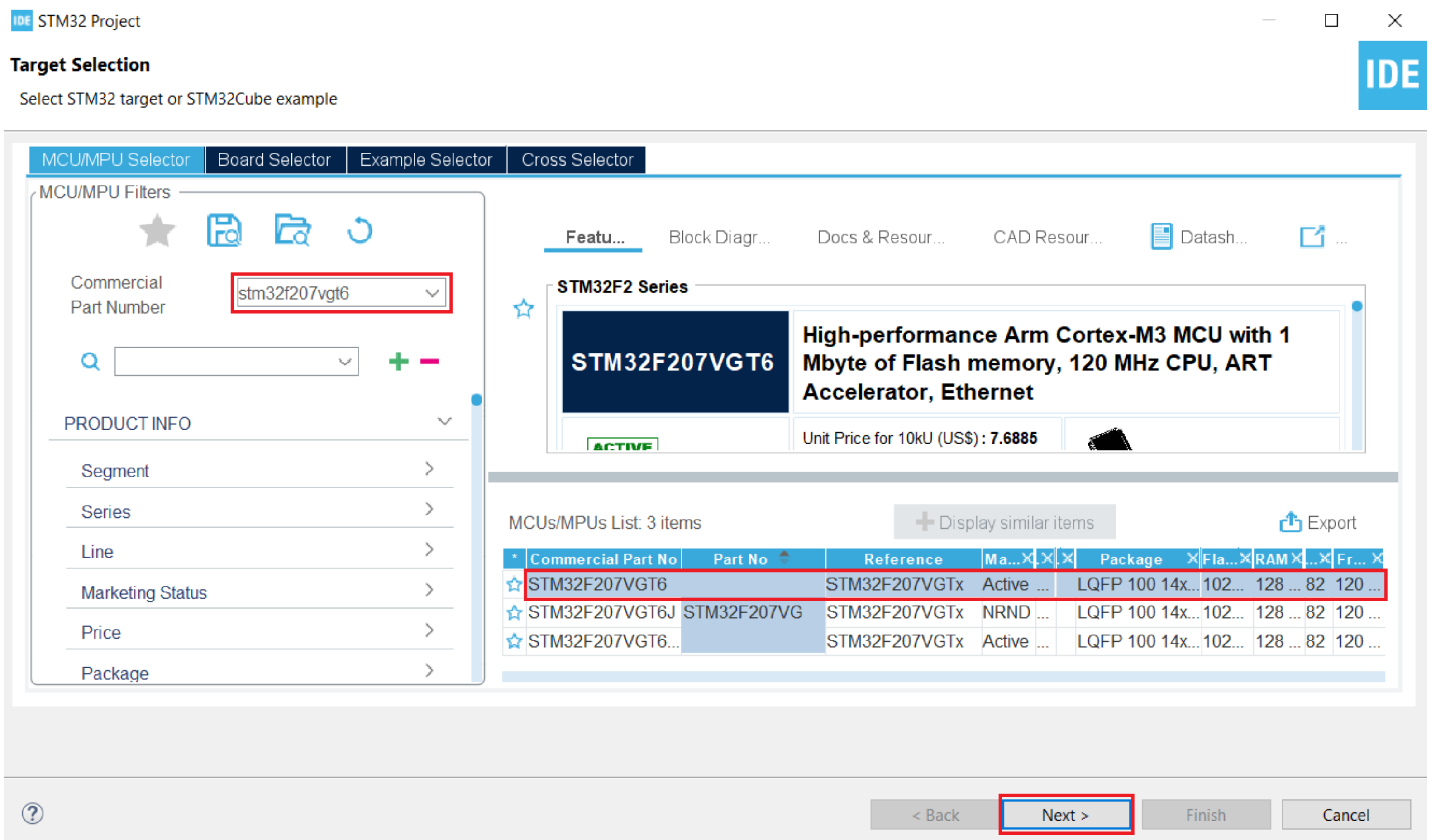
STM32CubeIDE 프로젝트 생성

- **"File" 클릭, "New" 클릭, "STM32 Project" 클릭.**



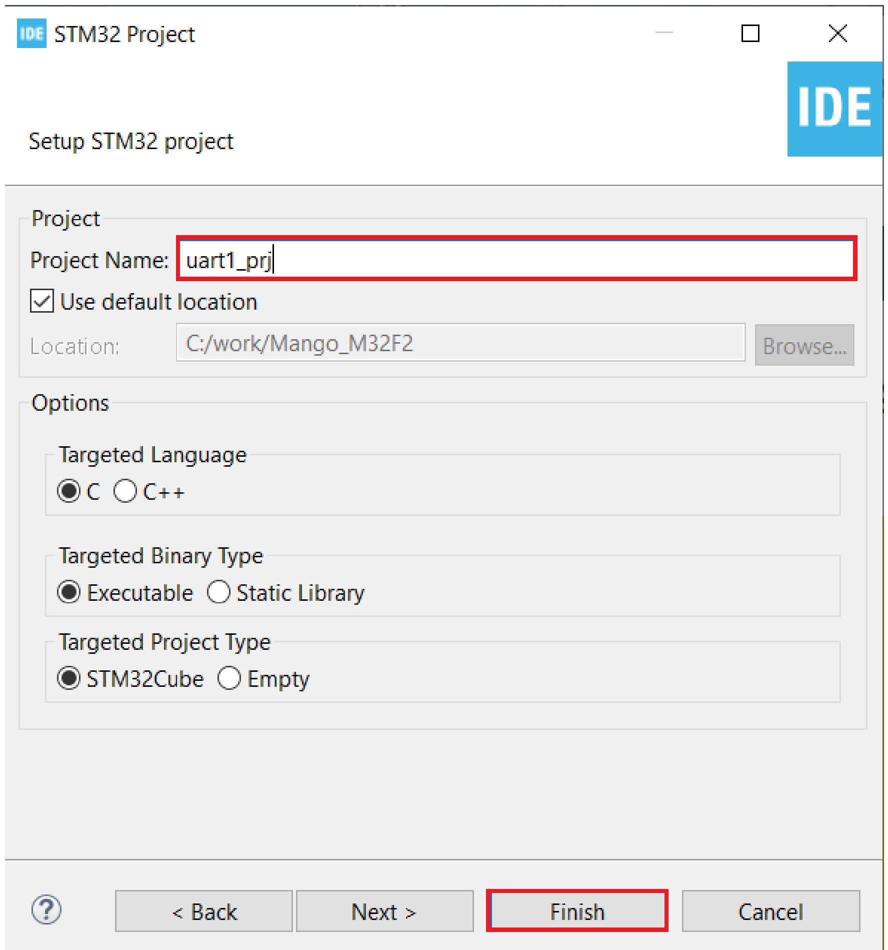
STM32CubeIDE 프로젝트 생성

“STM32F207VGT6“ 선택 후 "Next” 클릭

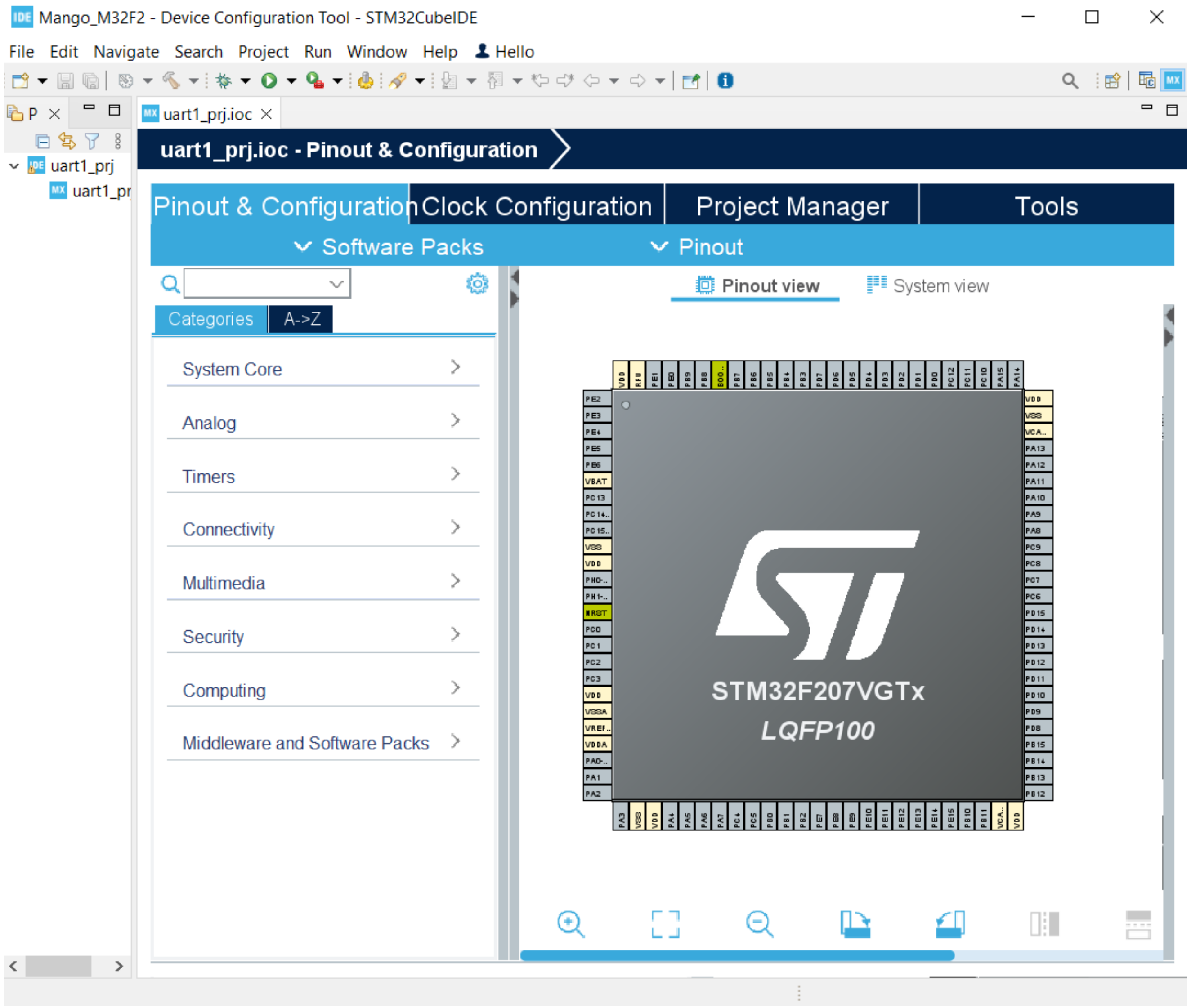


STM32CubeIDE 프로젝트 생성

- **Project Name으로 “uart1_prj” 입력 후 “Finish” 클릭.**



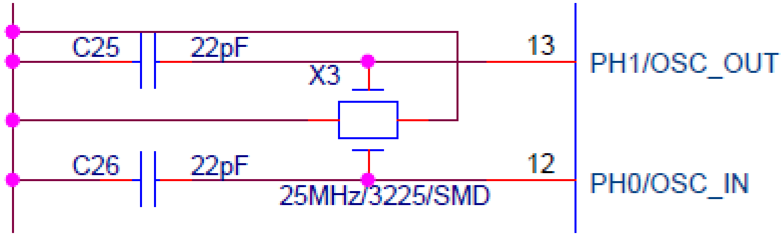
Pinout view



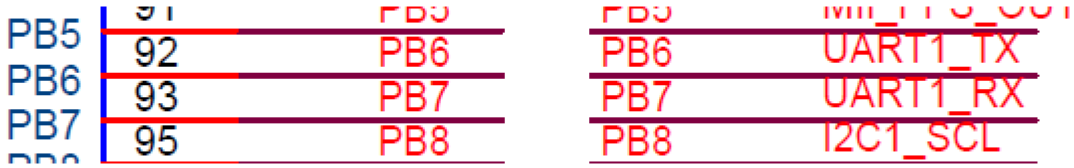
Mango_M32F2 - Device Configuration Tool - STM32CubeIDE
File Edit Navigate Search Project Run Window Help Hello
uart1_prj.ioc
uart1_prj.ioc - Pinout & Configuration
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Pinout view
System view
Categories
A->Z
System Core
Analog
Timers
Connectivity
Multimedia
Security
Computing
Middleware and Software Packs
STM32F207VGTx
LQFP100

Clock / UART1 pinmap

- **25MHz external clock - PH0 / PH1**

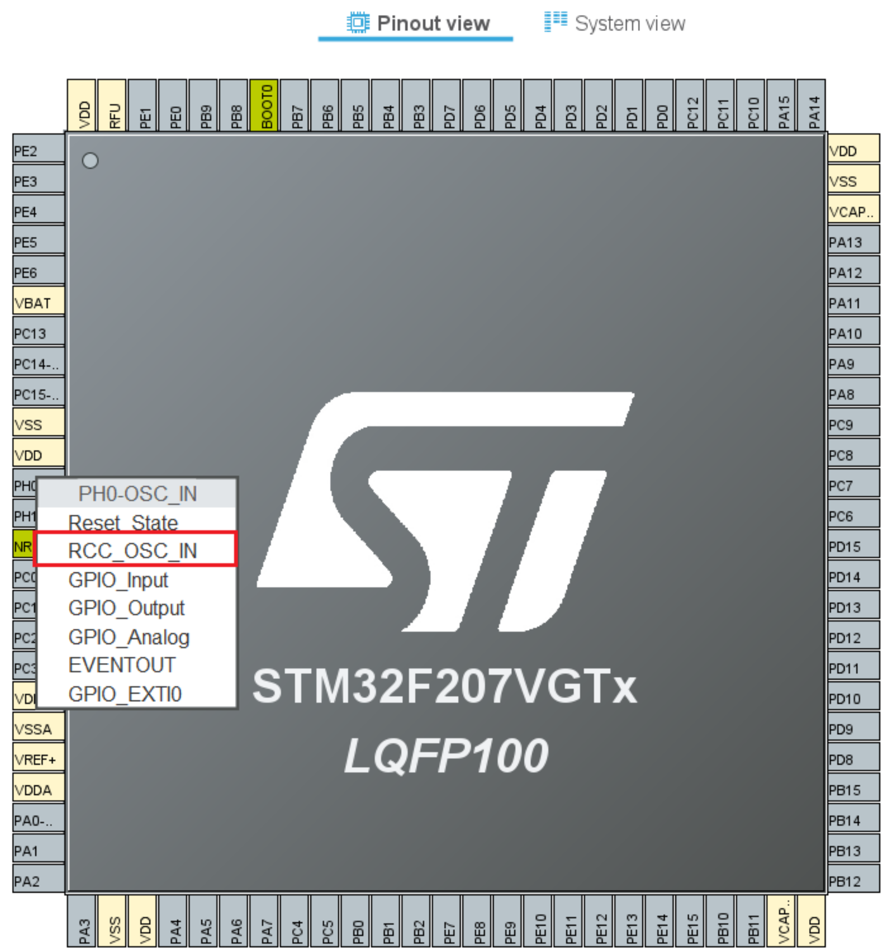


- **UART1 - PB6 / PB7**



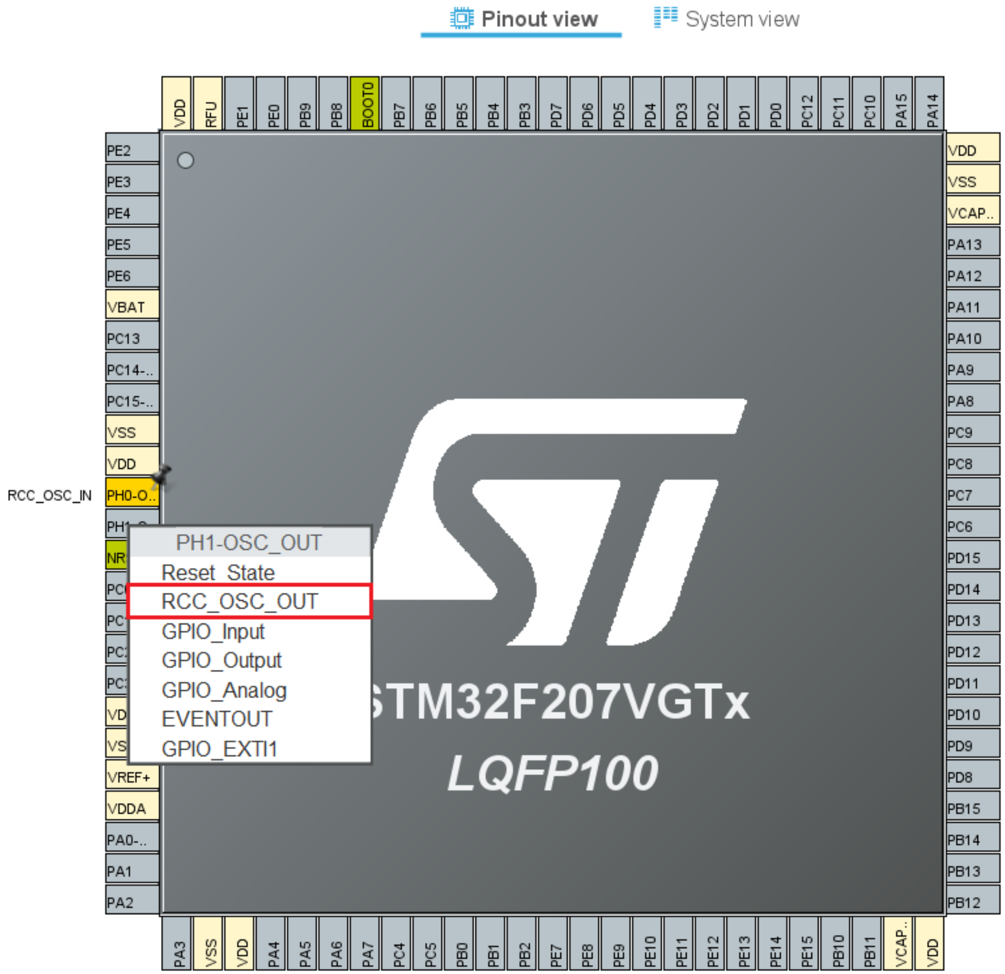
Pinout view

PH0 – RCC_OSC_IN 선택



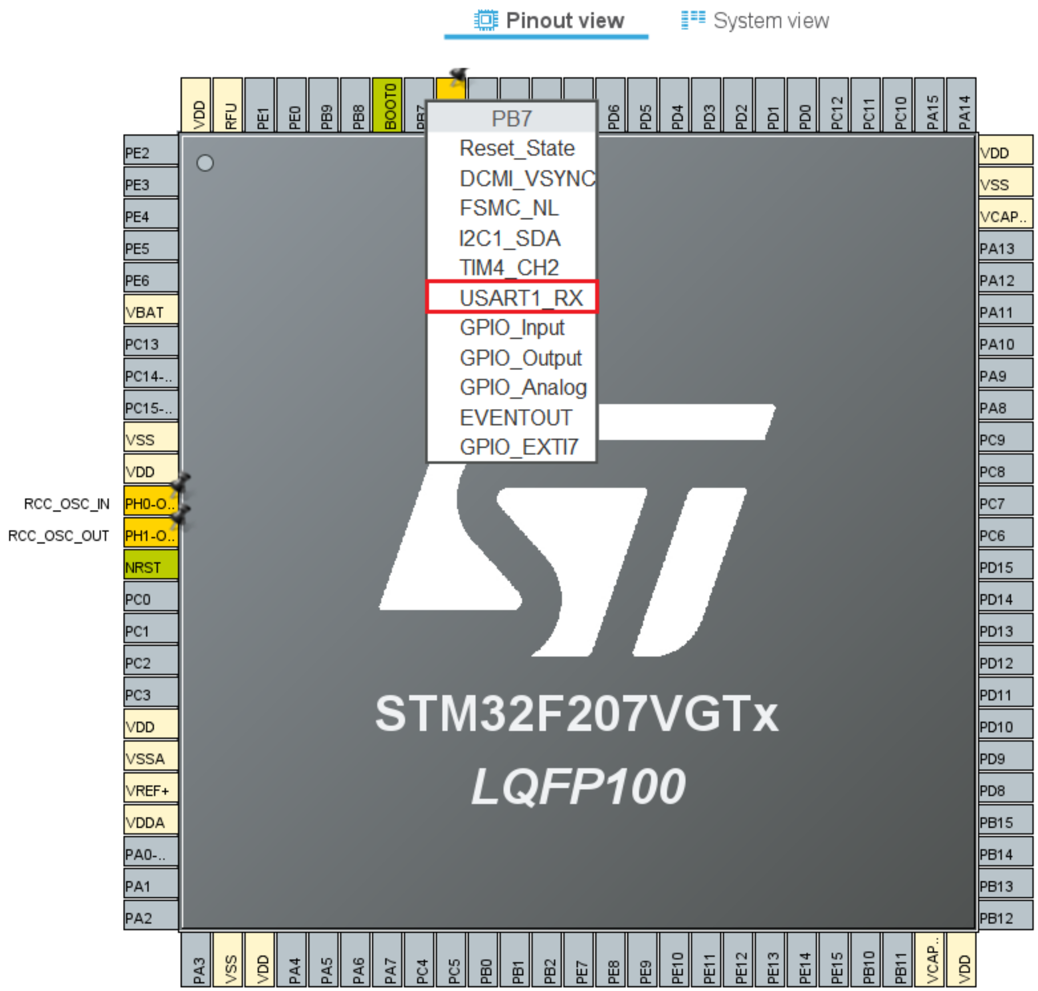
Pinout view

- **PH1 – RCC_OSC_OUT 선택**



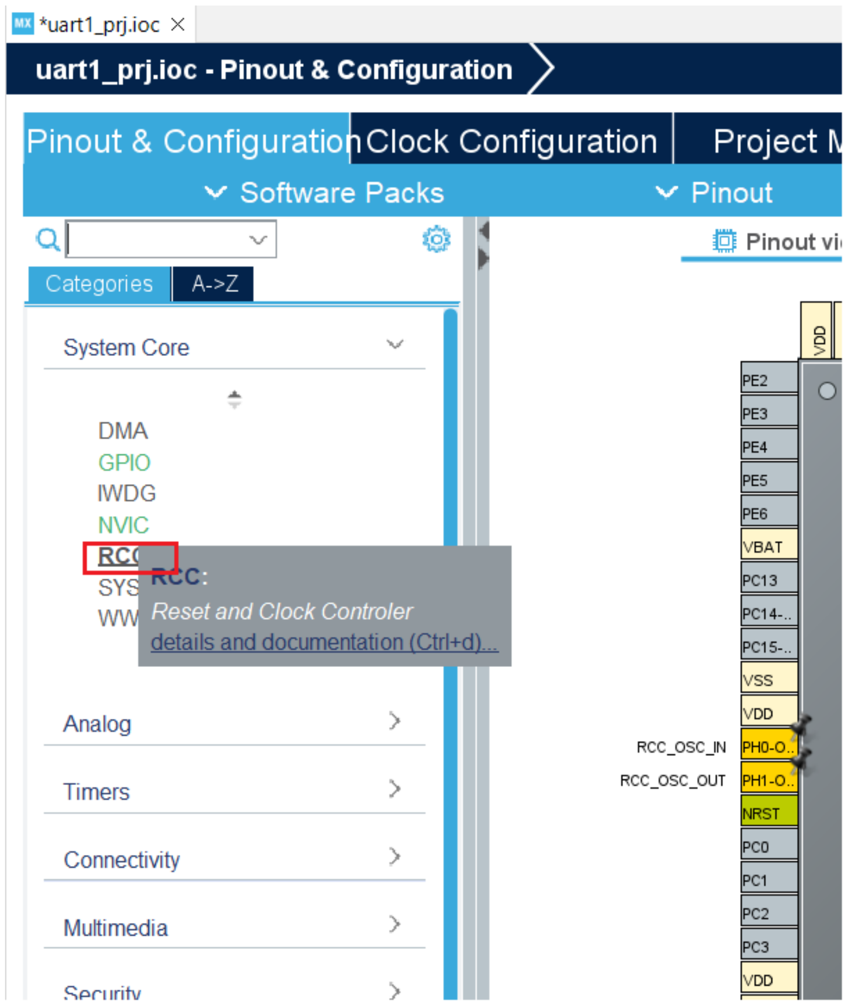
Pinout view

- **PB6 - USART1_TX, PB7 – USART1_RX 선택.**



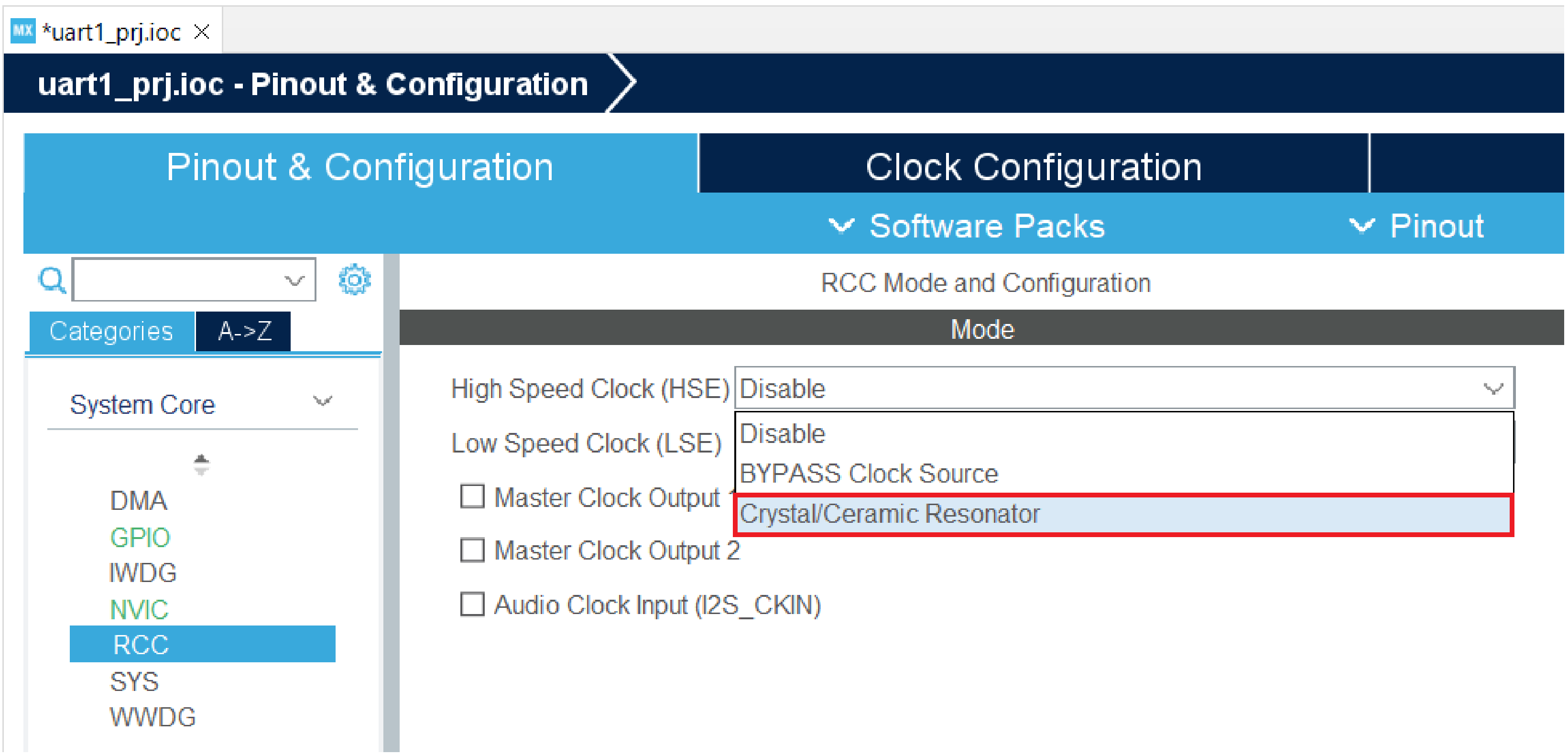
RCC 설정

■ **“System Core"에서 “RCC” 클릭.**



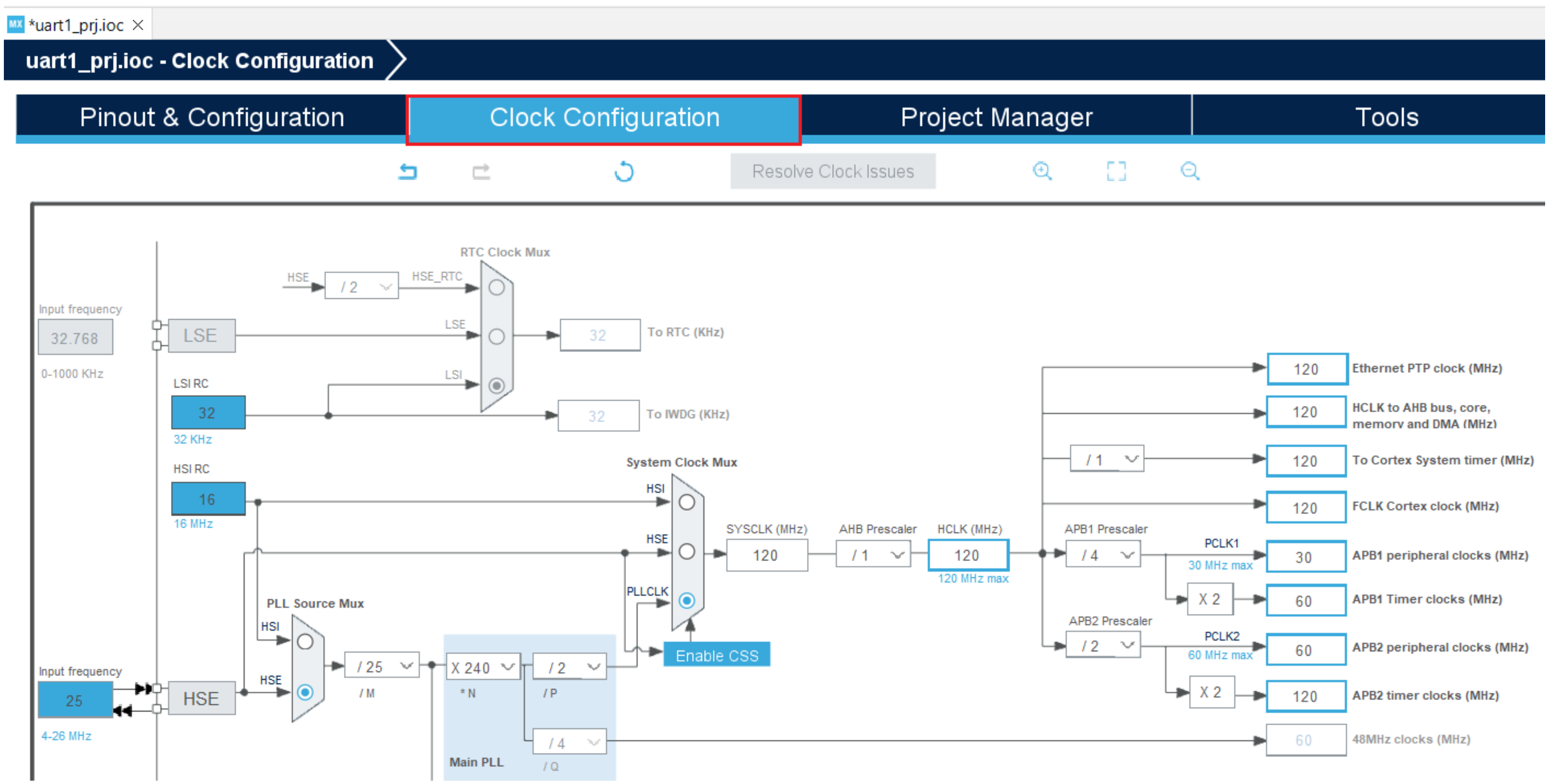
RCC 설정

■ **“High Speed Clock(HSE)"에서 “Crystal/Ceramic Resonator”클릭**



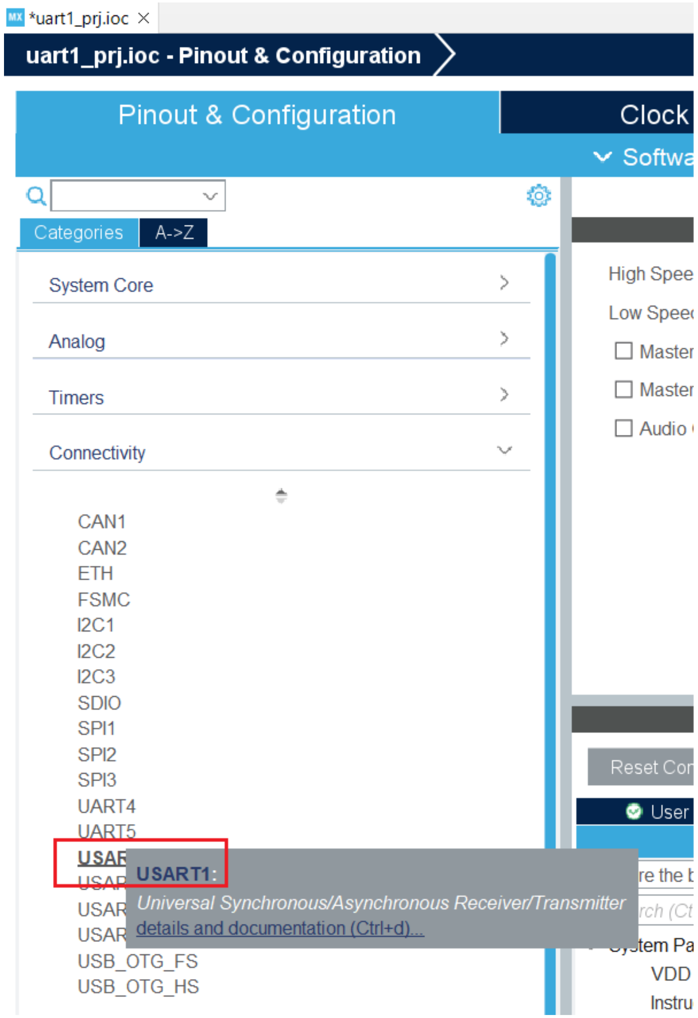
Clock Configuration

■ "Clock Configuration" 클릭 후 Clock Frequency 설정



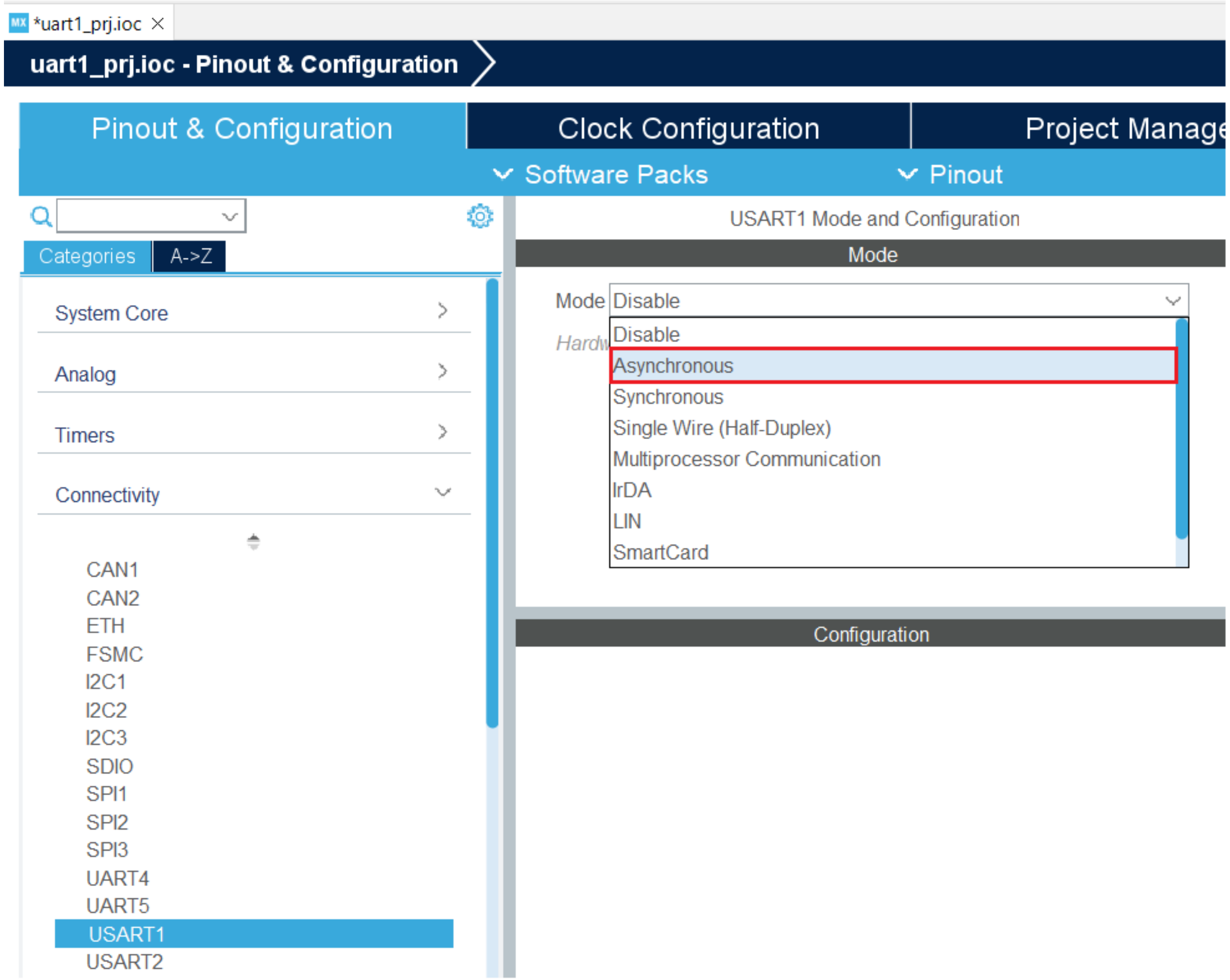
UART1 설정

- “Connectivity“ 클릭, “USART1” 클릭.



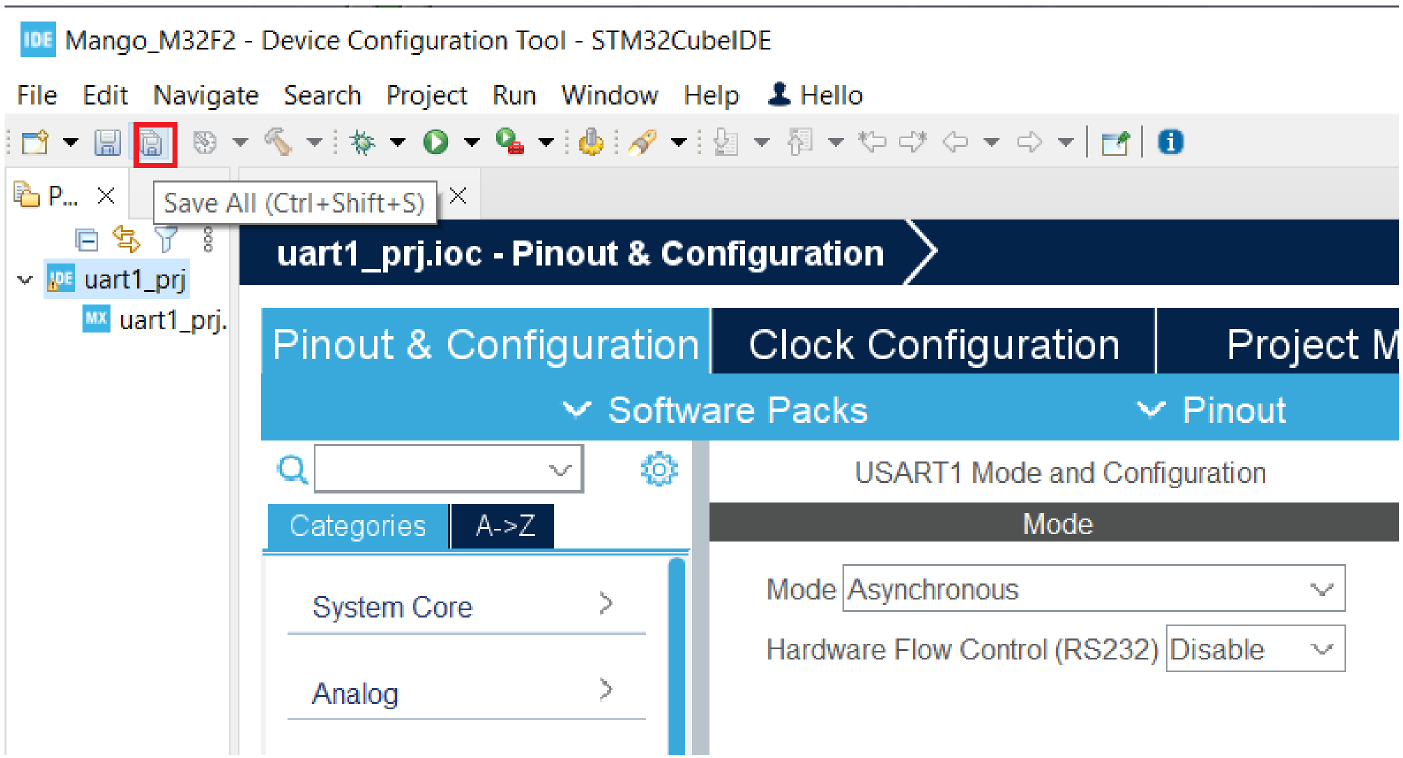
UART1 설정

- **“Mode“, “Asynchronous” 클릭.**



UART1 Test Project

- "Save All"클릭, Code Generation.



UART1 Test Project

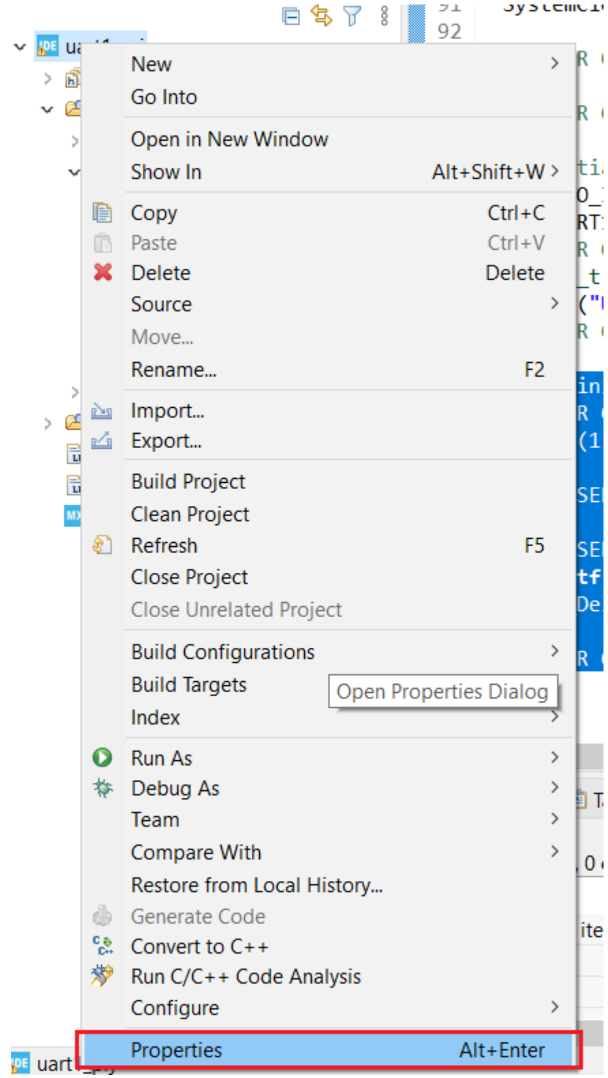
core/src/main.c 편집

- printf() – UART1, overwrite _write() function

```
/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
#define hUART huart1
int _write( int32_t file , uint8_t *ptr , int32_t len )
{
  /* Implement your write code here, this is used by puts and printf for example */
  for ( int16_t i = 0 ; i < len ; ++i )
  {
    HAL_UART_Transmit( &hUART, ptr++, 1, 100);
  }
  return len;
}
/* USER CODE END 0 */


/* USER CODE BEGIN 2 */
int count = 0;
printf("UART1 test\r\n");
/* USER CODE END 2 */


/* Infinite loop */
 /* USER CODE BEGIN WHILE */
 while (1)
 {
   printf("count=%d\r\n", count++);
   HAL_Delay(1000);
   /* USER CODE END WHILE */

   /* USER CODE BEGIN 3 */
}
 /* USER CODE END 3 */
```

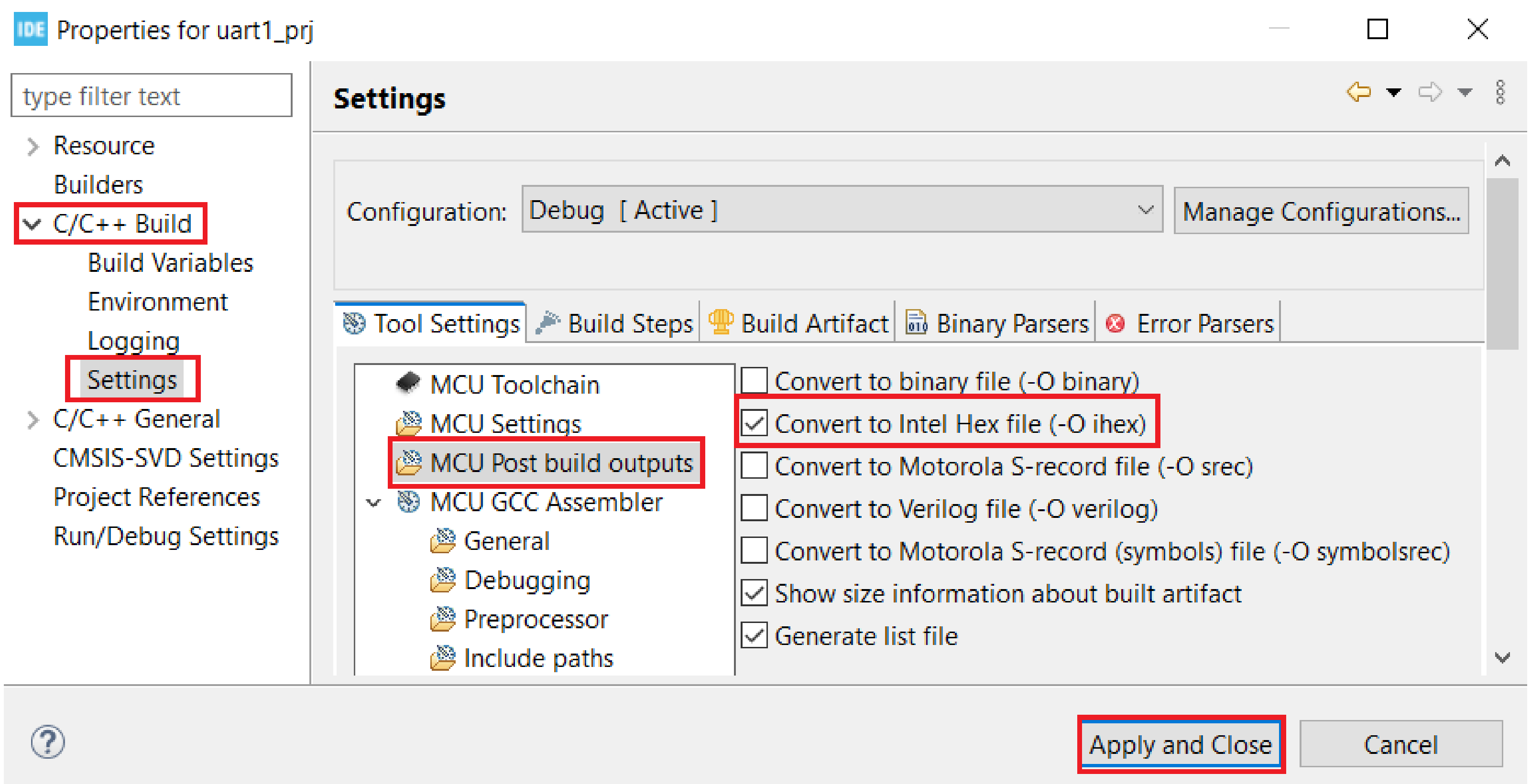
Hex File Generation

- “uart1_prj” 우클릭, “Properties“ 클릭.



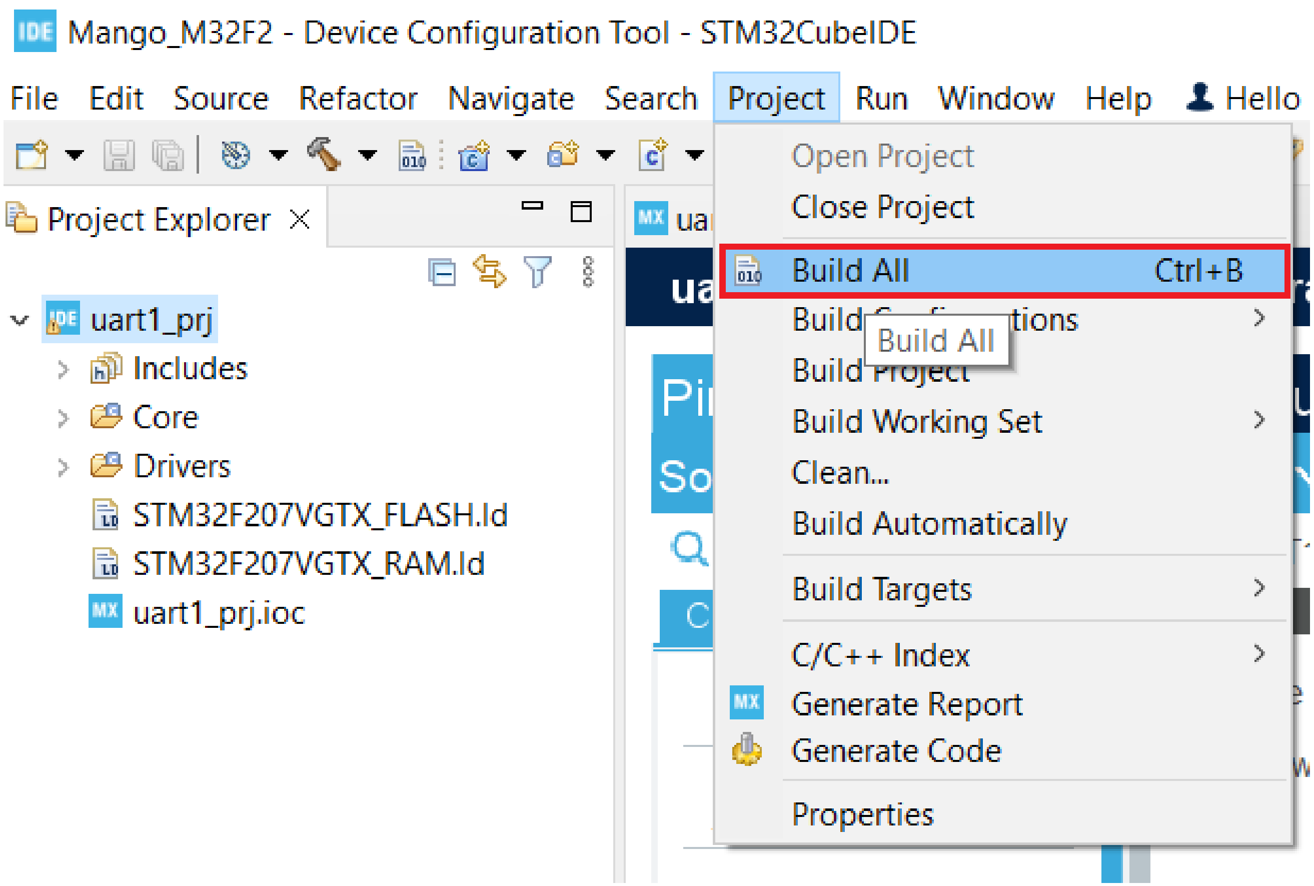
Hex File Generation

- "C/C++ Build"-"Settings"-"Tool Settings"-"MCU Post build outputs"-"Convert to Intel Hex file (-O ihex) 클릭,
- "Apply and Close" 클릭.



Project Build

■ “Project”클릭, “Build All“ 클릭.



Test 결과 확인

- 3핀 serial cable을 UART3에 연결하고 Flash Loader로 다운로드.
- 3핀 serial cable을 UART1에 연결하고 결과 확인.
- RCC / Clock Configuration / UART1 설정은 나머지 실습에 공통.

Thank You